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Application of marginal structural models for causal inference on the treatment effect for overall survival in randomized controlled trials with control arm patients switching to active intervention after disease progression

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ABSTRACT

This research explores the application of marginal structural models (MSMs) in evaluating the causal treatment effect of active intervention versus control on overall survival in randomized clinical trials (RCTs) allowing for control arm patients to switch to active intervention after disease progression. When MSMs are applied in RCTs under this type of treatment switching setting, the question of interest and model specifications differ from both observational studies and from RCTs where patients in both arms are permitted to take alternative treatments after disease progression. A violation of structural positivity may result as an undesired consequence if MSM model weights are constructed using data directly from both arms. This research proposes a two-step approach to avoid this issue. Through simulation studies, it is demonstrated that the proposed approach allows for MSM to be used for analyzing survival data to detect causal active treatment effects under this one-way treatment switching setting. Additionally, estimation for the causal effect of the active intervention as the next line (post-disease progression) therapy can also be obtained from the MSM approach. A case study is presented to illustrate the application of MSMs under this type of treatment switching setting.

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Inverse probability weighting; marginal structure models; one-way treatment switching; time-dependent confounding; treatment switching

1. Introduction

In phase 3 randomized controlled trials (RCTs), patients in the control arm are often permitted to switch to active intervention (hereafter referred to as “one-way treatment switching”) after disease progression due to ethical considerations. This type of trial is common for oncology indications. Under a larger trial design paradigm, one-way treatment switching is one type of intercurrent event (ICE). For RCTs with ICEs, the treatment policy strategy-based intent-to-treat (ITT) approach is often used in the primary analysis for time to overall survival (OS). This approach results in an estimand meant for measuring the total treatment regimen effect while ignoring ICEs, which may result in estimation of an active intervention benefit on OS that is different from what would have been observed without ICEs. Under a hypothetical strategy, the target causal effect of active intervention benefit on overall survival (hypothetical estimand) may be estimated per ICH E9(R1) (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials E9(R1) (2019).

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In the following, we describe a hypothetical strategy-based methodology for calculating the causal effect under the one-way treatment switching setting and support it with a case study, as well as thorough simulation studies. The causal active intervention benefit estimate on OS that would have been observed if there were no one-way treatment switching is helpful for regulatory approval evaluation and for health technology assessment (HTA) payer reimbursement evaluation.

In the past 20 years, multiple advanced statistical methods have been developed to estimate the causal treatment effect on OS adjusting for one-way treatment switching in RCTs under the hypothetical strategy, of which the following four are widely used in industry: inverse probability of censoring weighting (IPCW) (Rimawi and Hilsenbeck 2012; Robins and Finkelstein 2000; Rotnitzky and Robins 2005), rank-preserving structural failure time models (RPSFTM) (Robins and Tsiatis 1991), iterative parameter estimation (IPE) (Branson and Whitehead 2002), and two-stage estimation (TSE) (Latimer et al. 2017, 2019). Among these methods, RPSFTM, IPE and TSE employ survival time rescaling for control arm patients with treatment switching, while IPCW uses re-weighting. Each of these methods requires different assumptions and carries different strengths and limitations. For example, RPSFTM and IPE do not require the use of covariate information but need to meet the “common treatment effect” assumption. Both IPCW and TSE relax the “common treatment effect” assumption but rely upon the untestable assumption that there are no unmeasured confounders.

The application of a marginal structural model (MSM) (Robins 2000) is another approach that can be used to estimate the causal treatment effect of active intervention on OS while accounting for treatment switching in RCTs under the hypothetical strategy. MSMs have been extensively researched and widely used in longitudinal observational cohort studies to make causal inference where the effect of interest is confounded in a time-dependent manner through exposure change during follow-up. However, to date, the application of MSM under the RCT setting is not as popular as with observational cohort studies. In a systematic review by (Farmer et al. 2018), the authors state that “Further efforts may be needed to promote use of causal methods to address additional clinical questions within RCTs to maximize their value.” To promote awareness of and facilitate the use of MSM in RCTs under the hypothetical strategy, we recently published a manuscript on the application of MSM for causal inference in phase 3 oncology trials with treatment switching in both active intervention and control arms (Xu, Liu, and Wang 2022).

For RCTs having a one-way treatment switching design, we have the following two important observations: 1) only the control arm treatment effect is contaminated by treatment switching; and 2) patients randomized to active treatment are not representative of the same population of control patients who are also exposed to active treatment post-treatment switching, as these are relapsed patients with active exposure as next-line treatment. Therefore, when MSM is applied in RCTs with one-way treatment switching, model parametrization needs to be different from both cohort studies and RCTs where switching may occur in both control and active intervention arms to alternative therapies (hereafter referred to as “2-way treatment switching”).

This article addresses the issues related to the application of MSM in RCTs where control arm patients may switch to active intervention after disease progression. In Section 2, we discuss the MSM setting in RCTs with one-way treatment switching designs. In Section 3, we evaluate the operating characteristics as described in Section 2 through Monte Carlo simulation. In Section 4, we discuss reducing bias when applying the MSM for RCTs with one-way treatment switching designs. In section 5, we applied MSM to Takeda oncology phase III trial data as a case study. The conclusion and discussion follow in Section 6

2. Proposed MSM setting for RCTs with one-way treatment switching designs

In this article, we use $R = 0, 1$ to denote the randomized control and active intervention arms, respectively. A is used to denote that treatment switching has occurred, and $A(t)$ is used to denote switching treatment status at time t . A superscript T refers to treatment (as opposed to a superscript C , which refers to censoring), and a subscript T is used as a subject’s time of death after enrollment. V is a

vector of time-invariant covariates measured at baseline and $L(t)$ is a vector of time-dependent covariates. Note that $L(0) = V$. IPTW and IPCW stand for inverse-probability-treatment-weighting and inverse-probability-censoring-weighting, respectively. SW stands for stabilized weight. Lastly, $I(i)$ and $J(j)$ are used as subject and visit time indices, respectively.

Under the ITT principle, patients with treatment switching are considered as belonging to the control arm for the entire duration of their follow-up. As a result, the ITT estimand may underestimate the interventional benefit of the active treatment versus control because the randomized treatment effects can be confounded by the one-way treatment switching in the control arm. However, with MSM estimands, randomized and switched treatment effects can be estimated separately, allowing us to tease out the true effect of the active intervention treatment. To illustrate this, let us consider the following Cox proportional hazard model in which we model the conditional hazard of death at time t , $\lambda_T(t|\cdot)$ as:

$$\lambda_T(t|\bar{A}(t), R, V) = \lambda_0(t)e^{\{\gamma_0 V + \gamma_1 R + \gamma_2 \bar{A}(t)\}} \quad (1)$$

Note that $A(t)$ always equals zero for patients in the active treatment arm because in our setting they are not allowed to switch to the control arm, whereas for patients in the control arm, $A(t)$ equals one if the patient has switched to the active arm at time t and equals zero otherwise. An overbar represents covariate history. For example, $\bar{A}(t) = \{A(u); 0 \leq u < t\}$ represents patients' treatment switching history up to time t .

Model (1) assumes that the hazard of death at time t depends on the patient's R , V , and $A(t)$ history. In addition to $A(t)$, it is expected that there exist time-dependent $L(t)$ that change over time and affect the hazard of death, such as: relapse status, lab results, and biomarker values. It is easy to adjust for these time-varying covariates by simply adding $L(t)$ into Model (1). However, if $L(t)$ further satisfies the following two conditions: 1) it predicts both treatment switching and death, and 2) it is affected by treatment-switching treatment history; then, the contents of $L(t)$ become time-dependent confounders, in which case if we put $L(t)$ in Model (1) as time-dependent covariates, the estimate $\hat{\gamma}_2$ obtained by maximizing the Cox partial likelihood is an (asymptotically) unbiased estimate of the association parameter of γ_2 , but is a biased estimate of the causal effect of treatment switching therapy on OS (Hernán, M. Á., B. Brumback, and Robins 2000; Robins et al. 2000a), which also makes $\hat{\gamma}_1$ a biased estimate of the active intervention causal effect. To obtain unbiased causal estimates, $L(t)$ will appear in the weighting Equations (3), (4) and (4a), which will be discussed shortly, but will not be included in the time dependent Cox models, including Model (1).

MSM can be used to estimate the causal effect of a time-varying exposure in the presence of both fixed and time-dependent confounders. For RCTs with a one-way treatment switching design, patients in the active arm have zero probability of treatment switching. As will be further discussed in section 5.2, positivity exists when patients have a non-zero probability of undergoing or not undergoing treatment switching for every combination of confounder set values. When positivity, which is an assumption of MSM, is violated, bias will be introduced in the analysis (Branson and Whitehead 2002). Thus, for RCTs with a one-way treatment switching setting, if patients in both arms are included in estimating the causal effect, the positivity assumption will be violated structurally, i.e., it is a structural positivity violation. Therefore, to perform causal inference under the one-way treatment switching setting, we need to adopt the following two-step approach.

In the 1st step, we consider the randomized control arm data only, then Model (1) reduces to

$$\lambda_T(t|\bar{A}(t), R, V) = \lambda_0(t)e^{\{\gamma_0 V + \gamma_2 A(t)\}} \quad (2)$$

Model (2) has the same setting as an observational longitudinal cohort study, where control arm patients are allowed to switch to active intervention after disease progression. As a result, unless the proportion of treatment switching patients is very low (below 15%) or very high (above 85%), for which random positivity violations may incur, the probability of positivity violation is very low when performing causal inference. Thus, we may use MSM to estimate the causal effect for one-way

treatment switching by following the same procedures as in a cohort study via IPTW and IPCW estimators (Hernán, M. Á., B. Brumback, and Robins 2000), as depicted by Equations (3) and (4) below. Specifically, we first calculate the stabilized weights: $sw_{i,R=0}(t) = sw_{i,R=0}^T(t) * sw_{i,R=0}^C(t)$, where $sw_{i,R=0}^T(t)$ is the stabilized weight for treatment on control arm patients, defined as:

$$sw_{i,R=0}^T(t) = \prod_{j=1}^{int(t)} \frac{P[\bar{A}(j) = \bar{a}_i(j) | \bar{A}(j-1) = \bar{a}_i(j-1), V = v_i]}{P[\bar{A}(j) = \bar{a}_i(j) | \bar{A}(j-1) = \bar{a}_i(j-1), \bar{L}(j) = \bar{l}_i(j), V = v_i]}, \quad (3)$$

and $sw_{i,R=0}^C(t)$ is the stabilized weight for censoring on control arm patients, defined as:

$$sw_{i,R=0}^C(t) = \prod_{j=1}^{int(t)} \frac{P[C(j) = 0 | \bar{C}(j-1) = 0, \bar{A}(j-1) = \bar{a}_i(j-1), V = v_i]}{P[C(j) = 0 | \bar{C}(j-1) = 0, \bar{A}(j-1) = \bar{a}_i(j-1), \bar{L}(j) = \bar{l}_i(j), V = v_i]}. \quad (4)$$

In Equations (3) and (4), $int(t)$ is the total number of visits up to time t ; $\bar{A}(j)$ represents a vector of treatment switching history up to visit j ; $\bar{C}(j)$ represents a vector of censoring to accommodate informative censoring; and $\bar{L}(j)$ represents $L(t)$ history up to time t . We then use $sw_{i,R=0}(t)$ to re-weight each patient's contribution when fitting the time-dependent Cox model. The re-weighting effectively creates, for a risk set at time t , a pseudo-population in which: 1) $\bar{L}(j)$ no longer predicts initiation of treatment switching at time t , i.e., $\bar{L}(j)$ no longer contributes time-dependent confounding; 2) the causal association between switched therapy and survival is the same as in the original study source population (Hernán, M. Á., B. Brumback, and Robins 2000; Robins et al. 2000a). Of note that, even though Equations (3) and (4) look similar to the IPTW and IPCW equations used in MSM analyses for observational studies and RCTs with treatment switching, the involved patients are different: Equations (3) and (4) cover patients in the control arm only, while all patients are covered by the IPTW and IPCW equations when MSM is applied under the observational setting.

Based on the re-weighted pseudo-population, to estimate the causal effect of switched treatment within the control arm, we define a counterfactual random variable $T_{\bar{A}}$ to be a patient's time of death from randomization if the patient's treatment switching history had been $\bar{A}(t)$, rather than the observed history. We can observe $T_{\bar{A}}$ for those whose observed treatment switching history until the observed time of death agrees with $\bar{A}(t)$, i.e., $T_{\bar{A}} = T =$ actual time of death from randomization (Yamaguchi and Ohashi 2004). By applying this counterfactual result using the re-weighted population, Model (2) then becomes the following marginal structural Cox proportional hazards Model (5):

$$\lambda_{T_{\bar{A}}}(t | \bar{A}(t), R) = \lambda_0(t) e^{\{\beta_0 V + \beta_2 \bar{A}(t)\}}. \quad (5)$$

where $\lambda_{T_{\bar{A}}}(t | \cdot)$ is the hazard among patients in the control arm in the trial source population, had all patients followed switched-treatment through time t without any being right-censored. This is the structural model for the marginal distribution of the counter-factual variable $T_{\bar{A}}$. In model (5), β 's are estimated based on reweighted data from control arm patients only, where the stabilized treatment weights and stabilized censoring weights are derived via Equations (3) and (4), respectively. Based on Model (5), e^{β_2} has a causal interpretation as a hazard ratio at any time t , had all relapsed patients received active intervention as the next-line therapy, i.e. A, compared with the hazard had no relapsed patients received A at the time of randomization. For our RCT data analysis, Model (5) is not the primary interest, because the main goal of our MSM analysis is not to estimate the causal effect of A, the active therapy as next-line treatment for relapsed patients, but causal inference of R.

To address the primary question of interest, we proceed to Step 2 where randomized active arm data need to be added back into Model (5), which becomes:

$$\lambda(T_A)(t | \bar{A}(t), R) = \lambda_0(t) e^{\{\beta_0 V + \beta_1 R + \beta_2 \bar{A}(t)\}}. \quad (6)$$

In accordance with a one-way treatment switching design, patients in the active intervention arm have a zero probability of changing therapy. Therefore, the IPTW process should involve no patients in the $R = 1$ arm to avoid structural non-positivity. Since there is no treatment switching-introduced time-

dependent confounding, we can set $sw_{i,R=1}^T(t) \equiv 1$ at all observed time points for patient i with $R = 1$. However, informative censoring may still occur with patients in the active intervention arm, and thus the censoring-related $sw_{i,R=1}^C(t)$ weight still needs to be derived for patients in the $R = 1$ arm, defined as:

$$sw_{i,R=1}^C(t) = \prod_{j=1}^{int(t)} \frac{P[C(j) = 0 | \bar{C}(j-1) = 0, A(j-1) = a_i(j-1) = 1, V = v_i]}{P[C(j) = 0 | \bar{C}(j-1) = 0, A(j-1) = a_i(j-1) = 1, \bar{L}(j) = \bar{l}_i(j), V = v_i]} \cdot \quad (7)$$

In Model (6), β 's are estimated based on reweighted data from all patients, where the stabilized treatment weight is derived via Equation (3) for patients in the control arm and set as 1 for patients in active arm; the stabilized censoring weight is derived via Equations (4) and (7) for patients in the control and active arms, respectively. Then, we obtain $sw_i(t) = sw_{i,Re(0,1)}^T(t) * sw_{i,Re(0,1)}^C(t)$ for patients in both arms. For oncology phase 3 RCTs, stratified randomization is often used. Stratification variables may be included as part of V or as stratification factors in a stratified Cox model. With adequate sample size, and if randomization behaves as expected, baseline covariate distributions will be balanced between the active and control arms, in which case, as will be demonstrated by simulation results in Tables 1–4 in Section 3, V can be dropped in Model (6), i.e., Model (6) can be simplified as:

$$\lambda_{T_{\bar{A}}}(t | \bar{A}(t), R) = \lambda_0(t) e^{\{\beta_1 R + \beta_2 \bar{A}(t)\}}. \quad (8)$$

However, if V is not balanced at baseline and stabilized weights are applied, then V should be incorporated into Model (6) as will be demonstrated by simulation results in Tables 5 and 6 in Section 3

Following the above two steps, with Model (6) or (8), time-dependent confounding associated with treatment switching in the control arm and informative censoring in both arms on the OS outcome will be mitigated or eliminated through re-weighting. The estimate $\hat{\beta}_1$ obtained by maximizing the Cox partial likelihood is an (asymptotically) unbiased estimate of the causal effect of the log hazard ratio on OS for patients randomized to the active arm, compared to patients in the control arm without being confounded by treatment switching under the

Table 1. MSM, IPCW, and ITT model performance when both randomized and switched treatment effects are under the null distribution.

Simulation Parameters: $HR_R = 1(\beta_R = 0)$, $HR_A = 1(\beta_A = 0)$. Simulation iterations = 10000, $N = 300$ per arm, $\alpha = 0.05$; with balanced baseline covariate distribution between both arms

Mean Treatment Switching Rate (%)	Model Type	Mean HR_R	Absolute Bias (β_R)	Relative Bias (%) (β_R)	Type I Error (β_R)	95% CI Coverage (β_R)	SE ($\hat{\beta}_R$)
32.6	ITT	0.8516	−0.1484	−14.84	0.2028	0.7972	0.1280
	MSM	1.0096	0.0096	0.96	0.0539	0.9461	0.1342
	IPCW	1.0099	0.0099	0.988	0.0515	0.9485	0.1341
54.4	ITT	0.7586	−0.2414	−24.14	0.6057	0.3943	0.1285
	MSM	1.0107	0.0107	1.066	0.0520	0.9480	0.1534
	IPCW	1.0113	0.0113	1.125	0.0521	0.9479	0.1536
77.4	ITT	0.6725	−0.3274	−32.74	0.9015	0.0985	0.1294
	MSM	1.0262	0.0262	2.615	0.0577	0.9423	0.2032
	IPCW	1.0290	0.0290	2.899	0.0551	0.9449	0.2043
		Mean HR_A	Absolute Bias (β_A)	Relative Bias (%) (β_A)	Type I Error (β_A)	95% CI Coverage (β_A)	SE ($\hat{\beta}_A$)
32.6	MSM	1.0499	0.0499	4.987	0.0637	0.9363	0.2291
55.4	MSM	1.0042	0.0042	0.424	0.0551	0.9449	0.1966
77.4	MSM	1.0486	0.0486	4.863	0.0515	0.9485	0.2172

counterfactual setting; $\hat{\beta}_2$ is an (asymptotically) unbiased estimate of the causal effect of the log hazard ratio at any time t , under the counterfactual setting, had all relapsed patients received active intervention at the time of randomization, compared with the hazard had all relapsed patients received no active intervention. It is imperative to note that when MSM is applied under the one-way treatment switching trial setting, no R by $\bar{A}(t)$ interaction term should be added in Model (6) or (8), because treatment switching only occurs in the control arm. This is an important difference compared to the MSM application under the 2-way treatment switching setting (Xu, Liu, and Wang 2022).

In summary, with the proposed two-step MSM setting for RCTs with one-way treatment switching, after step 1 the randomized but treatment-switching-confounded control arm has been transformed into two arms: a re-weighted control arm and a re-weighted active intervention arm for relapsed patients in Model (5). After Step 2, the randomized re-weighted active intervention arm is incorporated into Model (6) or (8). The counting of time until death or right-censoring starts at randomization in all three arms (counterfactually for the active intervention arm for relapsed patients, i.e., treatment-switched arm). By comparing with the re-weighted control arm, we can obtain two estimates of the causal effects on OS: One for randomized active intervention and one for active intervention as next-line treatment on relapsed patients. The former is the primary question of interest to sponsors; the latter is a MSM by-product. In contrast to the RPSFTM (Robins and Tsiatis 1991) or IPE approaches (Branson and Whitehead 2002), MSM does not assume a “common treatment effect” for these two causal estimates.

3. Simulation study: algorithm and results

3.1. Simulation scope and datasets

To verify the statistical properties for the application of MSM in RCTs under a one-way treatment switching setting as described in Section 2, Monte Carlo simulations were conducted. In addition, simulations were also conducted for IPCW and ITT models to compare their performance versus MSM.

In simulations, a counting process dataset was generated in each iteration, resulting in each patient having multiple rows of data. Each observation has multiple columns of data including patient id, treatment group, fixed baseline covariate values, time-dependent confounding covariate values, disease progression and treatment switching status (for the active arm, treatment switching status is always 0), and survival and censoring status.

3.2. Simulation algorithm

To generate time-dependent confounding data for MSM and IPCW, the simulation algorithm by Young et al (Young and Tchetgen Tchetgen 2014) under the observational study setting was used and modified, where randomized treatment status (R) is generated in Step 1 at baseline. Steps 2 through 5 are used to generate time-dependent treatment switching status for the $R = 0$ arm only. In the following, A_m represents treatment switching status at cycle m ; $A(t)$ represents treatment switching status at time t ; and $\bar{A}(t) = \{A(u); 0 \leq u < t\}$ represents patients' treatment switching treatment history up to time t .

Specifically, for patient i at cycle m (assuming a 30-day cycle):

Step 1. At baseline, $m = 0$, and the randomized treatment arm $R = 0, 1$:

- Generate baseline covariates, such as $x_{1,i} \sim N(\mu_R, \sigma^2)$, and $x_{2,i} \sim \text{Bin}(p_R)$
- Generate baseline survival time $T_{0,i} \sim \exp(\lambda_0)$, without covariate or treatment effects.
- With treatment and covariate effects included, $T_i = 30 \times \text{MAX}[1, (T_{0,i}/\lambda_0)] \times e^{-\beta_1 * R} + a \times x_{1,i} + b \times x_{2,i}$ is the survival time without $A(t)$, where x_1 and x_2 are the baseline

covariates distributed as defined above; a and b are constants; define $L_0 = A_0 = Y_0 = 0$, and $\beta_1 = \log(\lambda_1/\lambda_0) = \log(HR)_R$

Step 2. Post-baseline, $R = 0, 1$, and $m = 1, 2, \dots, K$ (assuming data is administratively censored after K cycles). Generate the time-dependent confounder $L_{m,i}$ from some choice of $f(L_m|A_{m-1}, \bar{L}_{m-1}, X_1, X_2, Y_0, Y_m = 0|\theta)$, where θ 's are the regression parameters in the regression model and Y_0 is a fixed parameter (see step 3 for utility of Y_0). In this algorithm, $f(\cdot)$ is specified as a linear function of $(A_{m-1}, \bar{L}_{m-1}, X_1, X_2)$, i.e., regard $L_{m,i}$ as a time-varying biomarker that is affected by treatment switching and covariate history. The switching rate can be adjusted by Y_0 in $f(\cdot)$.

Step 3. $R = 0$, assuming that A is initiated right after disease progression (PD). Generate $A \sim \text{Bin}(p_a)$, with $p_a = \text{Sigmoid}\{\text{Logit}[P(\cdot)]\}$, and $P(\cdot) = \text{logit}[P(A_m = 1|A_m, \bar{L}_m, Y_m = 0)] = \alpha_0 + \alpha_1 \bar{L}_{m,i} + \alpha_2 A_{m-1,i}$. The higher the past and current L values, the higher probability $\Pr(A_t) = \Pr(PD)$. Thus, A is determined by past treatment and covariate history.

Step 4. $R = 0$, evaluate Y_m at previously generated $(\bar{A}_{m-1,i}, \bar{L}_{m-1,i})$:

- If $T_i > (m - 1) \times 30$, then $Y_{m,i} = 0$ and patient i is still alive by cycle $m < K$
- If $A_{m-1,i} = 1$, update survival time $T_i = (T_i - 30 \times (m - 1))e^{-\beta_2 \bar{R}_{m-1,i}} + 30(m - 1) + c\bar{L}_{m-1,i}$, where β_2 is $\log(HR)_A$ and c is a constant
- Go to $m + 1$ iteration for patient i
- Otherwise, $Y_{m,i} = 1$, i.e., patient i died by cycle m , with event time T_i
- Clearly, survival time is also impacted by treatment switching experience $(\bar{A}(t))$ and covariate history.

Step 5. $R = 0, 1$, if $T_i > (K \times 30)$, patient i is administratively censored; separately, generate independent censoring via uniform distribution at a 5% rate while $T_i \leq (K \times 30)$

For each simulation scenario, the randomization ratio was 1:1 and the administrative censoring rate was set at around 10%. The treatment switching distribution in the control arm was determined via the baseline survival distribution through λ_0 in Step 1, study cycle K in Step 2, and through Y_0 in Steps 2 and 3. By increasing the fixed parameter Y_0 value, we increase the treatment switching rate in the simulation runs. By using this fixed parameter, it may take a small number of simulation runs to achieve the target treatment switching rate through trial-and-error. The mean treatment switching rate was calculated by averaging the treatment switching proportion across simulation runs. Each scenario was simulated under a low ($\sim 30\%$), medium ($\sim 50\%$) or high ($\sim 70\%$) treatment switching rate. For some scenarios, extremely low ($\sim 15\%$) or extremely high ($\sim 85\%$) treatment switching rates were also simulated. The evaluation metrics are as follows: 1) absolute bias = true HR – mean estimated HR; 2) relative bias = (absolute bias/true HR) $\times 100\%$; 3) type I error (under the null) or power (under the alternative); 4) coverage ratio = the proportion of 95% CIs that cover the true HR; and 5) standard error of parameter estimates.

Simulated data were analyzed in SAS 9.4 using extended Cox models in the time-dependent covariate setting via counting process data input (SAS Institute Inc., 2011). Treatment switching and censoring probabilities were obtained by Cox regression approximated via pooled logistic regression models (D'Agostino et al. 1990) where SWs for MSM were calculated via expressions (3) and (4). Within a randomized experimental arm, SW values for treatment switching were set to one for patients at all time points, and censoring probabilities were obtained by Cox regression approximated via pooled logistic regression models via expression (4a). For ITT analyses, model-based standard errors for the β estimates were used. For the MSM and IPCW analyses, robust empirical standard errors for the β estimates were used.

3.3. Simulation results

In this section, we first summarize simulated results for the following four scenarios: 1) both the randomized active intervention and switched-treatment effects are under the null distribution ($HR_R = 1$ and $HR_A = 1$); 2) the randomized active intervention treatment effect is under the null distribution ($HR_R = 1$) and the switched-treatment is protective ($HR_A < 1$); 3) the randomized active intervention treatment is protective ($HR_R < 1$), and the switched-treatment effect is under the null distribution ($HR_A = 1$); 4) both the randomized active intervention treatment and switched-treatment are protective, but with different treatment effects. In these four scenarios, to verify that baseline covariates in the MSM model may be dropped when they are balanced between the active and control arms and SW is used, we generated baseline covariates under the same distributions for both arms and then dropped them from the MSM analysis. For a fair comparison with MSM, baseline covariates were also dropped from the IPCW and ITT models. In the 5th scenario, baseline covariates were generated under different distributions in active and control arms. Their impact was tested by either including or excluding them in the MSM and IPCW models with SWs.

As discussed in Section 2, the proposed MSM approach does not assume that $HR_R = HR_A$ (i.e., $\beta_R = \beta_A$). The switched-treatment effect β_A can be estimated and used for inference via a hypothetical MSM estimand as a byproduct of Model (6) or (8) in Section 2. On the other hand, in theory and practice, $A(t)$ is never included in ITT or IPCW models.

Tables 1–4 summarize simulation results from scenarios 1 to 4. Sample sizes tested under different simulation scenarios ranged from $N = 150$ – 600 per arm. The simulations demonstrated that the impact of sample size on the performance metrics (except for power) is limited. Therefore, simulation results are presented for a sample size of 300 per arm only. Simulation results for $N = 150$ and 600 per arm are summarized in the supplementary material. Tables 5 and 6 summarize simulation results for scenario 5 with baseline covariates included and excluded from the MSM and IPCW models. Tables 1–4 include simulation results with low, medium, and high treatment switching rates only. Tables 5 and 6 also include simulation results with an extremely high treatment switching rate.

Table 1 summarizes MSM performance when both randomized and switched-treatment effects are under the null distribution. For the randomized treatment effect, both MSM and IPCW yield empirical type I error rates close to the nominal (0.05) level, but the empirical type I error is highly inflated with the ITT approach, which also yields biased HR estimates. Relative bias is consistently less than 3% percent with MSM and IPCW; but for ITT models, higher treatment switching rates lead to more biased HR estimates. The coverage ratio is close to the 95% nominal level for MSM and IPCW when the treatment switching rates are not too low or too high; but is very poor for ITT models where there is evidence that it worsens with increasing treatment switching rates; however, the standard error for the β_R estimates is larger for MSM and IPCW than ITT, as more variability is introduced during the re-weighting process. For the switched treatment effect, with MSM, the empirical type I error is close to the nominal level of 0.05, the coverage is close to 95%, and the relative bias is less than 5% under different treatment switching rates.

Table 2 summarizes MSM performance when the randomized active intervention treatment effect is under the null distribution and treatment switching is protective. MSM and IPCW both yield an empirical type I error around the nominal level (0.05) for testing the randomized treatment effect when the treatment switching rate ranges from approximately 30% to around 77%; in contrast, the empirical type I error is very high for ITT. The relative bias is consistently less than 3% for MSM but is very high for ITT. The standard error for β_R estimates is larger for MSM than ITT. When the switched-treatment effect is under the alternative, the coverage ratio is around the 95% nominal level and the relative bias is less than 3% under different treatment switching rates for MSM, but higher treatment switching rates lead to increased bias.

Table 3 summarizes MSM performance when the randomized active intervention treatment is protective, and the switched-treatment effect is under the null distribution. Both MSM and IPCW have good performance for estimation and inference on the randomized treatment effects: the

Table 2. MSM, IPCW, and ITT model performance when the randomized active intervention treatment effect is under the null distribution and switched treatment is protective.

Simulation Parameters: $HR_R = 1(\beta_R = 0)$, $HR_A < 1(\beta_A < 0)$. Simulation iterations = 1000, $N = 300$ per arm, $\alpha = 0.05$; with balanced baseline covariate distribution between both arms

Mean Treatment Switching Rate (%)	Model Type	Mean HR_R	Absolute Bias (β_R)	Relative Bias (%) (β_R)	Type I Error (β_R)	95% CI Coverage (β_R)	SE $(\hat{\beta}_R)$
32.7	ITT	0.855	−0.145	−14.53	0.197	0.803	0.128
	MSM	1.0024	0.0024	0.241	0.049	0.951	0.134
	IPCW	1.0035	0.0035	0.352	0.045	0.955	0.134
54.5	ITT	0.767	−0.233	−23.26	0.574	0.426	0.129
	MSM	1.0065	0.007	0.654	0.043	0.957	0.154
	IPCW	1.0137	0.0137	1.365	0.042	0.958	0.153
77.7	ITT	0.682	−0.318	−31.76	0.880	0.120	0.130
	MSM	1.012	0.012	1.232	0.044	0.956	0.205
	IPCW	1.033	0.033	3.335	0.041	0.959	0.204
		True HRA	Mean HRA	Absolute Bias (β_A)	Relative Bias (%) (β_A)	95% CI Coverage (β_A)	SE $(\hat{\beta}_A)$
32.7	MSM	0.819	0.823	0.0047	0.579	0.945	0.232
54.4	MSM	0.819	0.804	−0.015	−1.463	0.948	0.197
77.7	MSM	0.819	0.797	−0.022	−2.687	0.946	0.219

Table 3. MSM, IPCW, and ITT model performance when the randomized active intervention treatment is protective, and the switched treatment effect is under the null distribution.

Simulation Parameters: $HR_R < 1(\beta_R < 0)$, $HR_A = 1(\beta_A = 0)$. Simulation iterations = 1,000, $N = 300$ per arm, $\alpha = 0.05$; with balanced baseline covariate distribution between both arms

Mean Treatment Switching Rate (%)	Model Type	True HR_R	Mean HR_R	Absolute Bias (β_R)	Relative Bias (%) (β_R)	95% CI Coverage (β_R)	Power $(\hat{\beta}_R)$
29.8	ITT	0.700	0.973	0.273	39.0	0.072	0.055
	MSM	0.700	0.701	.0010	0.147	0.948	0.952
	IPCW	0.700	0.701	.0013	1.812	0.953	0.953
49.9	ITT	0.700	0.954	0.254	36.31	0.107	0.068
	MSM	0.700	0.706	0.006	0.855	0.954	0.876
	IPCW	0.700	0.706	0.006	0.893	0.953	0.875
69.8	ITT	0.700	0.939	0.239	34.16	0.101	0.101
	MSM	0.700	0.704	0.004	0.596	0.941	0.711
	IPCW	0.700	0.705	0.004	0.636	0.942	0.694
		Mean HRA	Absolute Bias (β_A)	Relative Bias (%) (β_A)	Type I Error (β_A)	95% CI Coverage (β_A)	SE $(\hat{\beta}_A)$
29.8	MSM	1.019	0.019	1.945	0.058	0.942	0.164
49.9	MSM	1.013	0.013	1.257	0.049	0.951	0.140
69.8	MSM	1.010	0.010	0.998	0.062	0.938	0.155

coverage ratio is around the 95% nominal level and the relative bias is less than 2%. ITT performance, on the other hand, is poor: the relative bias is around 35%, and the coverage is around 10% with all tested treatment switching rates. As expected, for both MSM and IPCW, power for detecting the randomized treatment protective effect decreases along with an increase in the proportion of relapsed patients in the control arm switching to an active agent. The decrease is faster for IPCW than MSM. When the switched-treatment effect is under the null, the empirical type I error and coverage ratio is around nominal level for MSM, and the relative bias is less than 2% under the tested treatment switching rates.

Table 4. MSM, IPCW, and ITT model performance when both the randomized active intervention treatment and switched treatment are protective, but with different treatment effects.

Simulation Parameters: $HR_R = 0.7(\beta_R < 0)$, $HR_A = 0.835(\beta_A < 0)$. Simulation iterations = 1,000, $N=300$ per arm; with balanced baseline covariate distribution between both arms

Mean Treatment Switching Rate (%)	Model Type	Mean HR_R	Absolute Bias (β_R)	Relative Bias (%) (β_R)	95% CI Coverage (β_R)	Power (β_R)
29.7	ITT	0.977	0.277	39.58	0.060	0.037
	MSM	0.700	−.0000	−0.0031	0.947	0.957
	IPCW	0.701	0.0009	0.1327	0.950	0.953
50.0	ITT	0.956	0.256	36.62	0.094	0.064
	MSM	0.707	0.0067	0.9553	0.954	0.876
	IPCW	0.707	0.0068	0.9768	0.952	0.872
60.5	ITT	0.948	0.248	35.38	0.133	0.079
	MSM	0.704	0.0040	0.5676	0.950	0.822
	IPCW	0.704	0.0039	0.5581	0.957	0.811
		Mean HR_A	Absolute Bias (β_A)	Relative Bias (%) (β_A)	95% CI Coverage (β_A)	Power (β_A)
29.7	MSM	0.847	0.0122	1.464	0.943	0.186
50.0	MSM	0.848	0.0123	1.469	0.937	0.231
60.5	MSM	0.843	0.0079	0.9502	0.957	0.239

Table 5. MSM and IPCW performance under the null when baseline covariates are generated under different distributions in each arm.

Simulation Parameters: $HR_R = 1(\beta_R = 0)$, $HR_A = 1(\beta_A = 0)$. Simulation iterations = 1,000, $N = 300$ per arm; baseline covariates: $X_{1i} \sim N(5, 6)$, $X_{2i} \sim \text{Bin}(p = 0.3) \in R = 1$; and $X_{1i} \sim N(20, 6)$, $X_{2i} \sim \text{Bin}(p = 0.2) \in R=0$

Mean Treatment Switching Rate (%)	Model Type	(X_1, X_2) in the Model?	Mean HR_R	Type I Error (β_R)	Relative Bias (%) (β_R)	95% CI Coverage (β_R)
28.5	MSM	No	1.0709	0.107	7.0895	0.893
	MSM	Yes	1.0240	0.049	2.3994	0.951
	IPCW	No	1.0711	0.106	7.1127	0.894
	IPCW	Yes	1.0217	0.051	2.1719	0.949
45.99	MSM	No	1.0907	0.131	9.0712	0.869
	MSM	Yes	1.0420	0.035	4.2025	0.965
	IPCW	No	1.0916	0.131	9.1561	0.869
	IPCW	Yes	1.0104	0.036	3.1041	0.964
86.9	MSM	No	3.903	0.490	290.29	0.510
	MSM	Yes	2.174	0.320	117.35	0.680
	IPCW	No	1.8340	0.474	83.396	0.526
	IPCW	Yes	1.4698	0.211	46.998	0.789
		(X_1, X_2) in the Model?	Mean HR_A	Type I Error (β_A)	Relative Bias (%) (β_A)	95% CI Coverage (β_A)
28.5	MSM	No	1.0558	0.073	5.5812	0.927
	MSM	Yes	1.0363	0.060	3.6277	0.940
45.99	MSM	No	1.0678	0.084	6.7806	0.916
	MSM	Yes	1.0386	0.046	4.2338	0.954
86.9	MSM	No	4.6062	0.470	360.62	0.530
	MSM	Yes	2.5543	0.444	155.43	0.556

Table 4 summarizes MSM performance when both the randomized active intervention treatment and switched treatment are protective, but with different treatment effects. For the randomized treatment effect, relative bias is less than 1% and the coverage is close to 95% for both MSM and IPCW. ITT performance is poor above 30%. As expected, for both MSM and IPCW, power for detecting the randomized treatment protective effect decreases along with an

Table 6. MSM performance under the alternative when baseline covariates are generated under different distributions in each arm.

Simulation Parameters: $HR_R = 0.7(\beta_R = -0.36)$, $HR_A = 0.9(\beta_A = -0.1)$. Simulation iterations = 1,000, $N=300$ per arm; baseline covariates: $X_{1i} \sim N(5, 6)$, $X_{2i} \sim \text{Bin}(p = 0.3) \in R=1$; and $X_{1i} \sim N(20, 6)$, $X_{2i} \sim \text{Bin}(p = 0.2) \in R=0$

Mean Treatment Switching Rate (%)	Model Type	(X_1, X_2) in the Model?	Mean HR_R	Power (β_R)	Relative Bias (%) (β_R)	95% CI Coverage (β_R)
20.1	MSM	No	0.7379	0.923	5.4022	0.932
	MSM	Yes	0.7126	0.606	1.7944	0.945
	IPCW	No	0.7385	0.924	5.4947	0.928
	IPCW	Yes	0.7106	0.586	1.5147	0.963
48.1	MSM	No	0.7584	0.744	8.3399	0.894
	MSM	Yes	0.7361	0.429	5.1545	0.940
	IPCW	No	0.7594	0.731	8.4797	0.892
	IPCW	Yes	0.7350	0.394	4.9962	0.941
83.3	MSM	No	0.9037	0.097	20.369	0.791
	MSM	Yes	0.8905	0.107	27.199	0.853
	IPCW	No	0.9064	0.094	29.478	0.785
	IPCW	Yes	0.8673	0.116	23.894	0.890
		(X_1, X_2) in the Model?	Mean HR_A	Power (β_A)	Relative Bias (%) (β_A)	95% CI Coverage (β_A)
20.1	MSM	No	0.9486	0.068	4.8346	0.941
	MSM	Yes	0.9301	0.083	2.7936	0.945
48.1	MSM	No	0.9645	0.076	6.5948	0.919
	MSM	Yes	0.9467	0.429	4.6258	0.940
83.3	MSM	No	1.1608	0.081	28.283	0.818
	MSM	Yes	1.1648	0.104	28.729	0.798

increase in the proportion of relapsed patients and the coverage is close to 95% with MSM for the tested cases. The increase in power for the switched-treatment effect is limited by an increase in the proportion of relapsed patients in the control arm switching treatment under the simulation setting.

Table 5 summarizes MSM and IPCW performance when both the randomized active intervention treatment and switched-treatment effect are under the null. As expected, 1) when baseline covariates are not balanced between treatment arms and they are excluded in the MSM or IPCW model, while stabilized weights are used, their performance will not be acceptable; and 2) as will be discussed in Section 4, when the proportion of treatment switching is as extremely high ($>85\%$), MSM (and IPCW) performance will be affected by random non-positivity and will also be unacceptable. When the proportion of treatment switching is extremely low ($<15\%$), MSM (and IPCW) performance may become unacceptable as well (some simulation outputs are summarized in Table 5a and 6a in the supplementary material).

Table 6 summarizes MSM and IPCW performance when both the randomized active intervention treatment and one-way treatment switching treatment effect are under the alternative. The conclusions are the same as Table 5.

4. Reducing bias when applying MSM for RCTs with one-way treatment switching designs

When MSMs are applied to analyze data from observational studies, diagnostics and sensitivity analyses are essential for their proper use (Cole and Hernán 2008). The same careful practice needs to be exercised when MSM is used to analyze data from RCTs with one-way treatment switching, so that bias in causal inference is controlled. In the setting of an RCT, most variables of interest can and should be pre-specified in the statistical analysis plan (SAP).

4.1. Accommodating the unmeasured confounder and consistency assumptions

The assumption of no unmeasured confounding is not testable. In the RCT setting, to accommodate this assumption adequately, it is important to make sure that a rich collection of prognostic factors be collected at the trial planning stage – i.e., all major potential confounding variables, both baseline and time dependent, that may affect disease progression, treatment switching, and/or death should be identified in the trial design stage, included in the case report forms, and listed in the SAP. This does not imply that one should include as many variables as possible in the Cox regression approximated by the pooled logistic regression models (D’Agostino et al. 1990) for weight computing, because: 1) the addition of too many covariates relative to the number of observations may introduce finite sample bias due to non-positivity, and 2) if non-confounding variables are also included in the pooled logistic regression models for deriving weights, the efficiency of the effect estimates will be negatively affected. Therefore, the identification of major potential confounding variables needs to be pre-planned carefully, via feedback from clinicians and key opinion leaders (KOL’s), and based on intensive literature research. Then, these confounders need to be prespecified in the SAP as covariate candidates for the pooled logistic regression models.

The consistency assumption in causal inference implies that if a patient’s counterfactual exposure could hypothetically exist, it would be a viable exposure and it and its effect on the outcome could be defined without ambiguity (Hernán et al. 2001). The consistency assumption is rarely exactly true and is also difficult to verify. However, if a RCT’s execution closely follows the study protocol, consistency is a reasonable assumption as it is not difficult to picture a scenario where a patient is hypothetically randomized to a different treatment than what was observed in reality (Hernán et al. 2001). Alternatively, in an observational study, the counterfactual for exposures such as weight are difficult to understand hypothetically and, depending on how they are defined, may have a variety of different forms and effects on the outcome. However, RCTs are not immune to violations of consistency. For example, if there is widespread exposure misclassification or non-adherence to randomized treatment in a RCT, the consistency assumption may no longer be plausible. For such cases, ad-hoc sensitivity analyses need to be conducted to investigate departure from the consistency assumption and the associated bias introduced in the MSM causal inference outcome.

4.2. Accommodating the positivity assumption

For RCT’s with treatment switching, positivity is defined as the assumption that patients have a non-zero probability of receiving or not receiving alternative therapy at every level of the combination of confounder set values. When the positivity assumption is violated, bias will be introduced in the MSM causal inference (Cole and Hernán 2008; Robins et al. 2000a). As elaborated on in Section 2, for RCTs with a one-way switching design, where only patients in the control arm are allowed to switch to the active arm after disease progression, if MSM is executed without using a two-step approach, any causal inference will be biased because of structural non-positivity.

Besides structural zeros, random zeros may also introduce non-positivity (Cole and Hernán 2008), which usually occurs by chance, with no patients exposed (or not exposed) to treatment switching for certain sets of exposure and covariate history. In RCTs with one-way treatment switching designs, when MSM is applied to analyze confounded OS data, the likelihood of random violations increases mainly owing to the following three reasons: 1) the treatment switching rate is too low or too high in the control arm, 2) too many weak confounding variables are included in the pooled logistic regression model for inverse probability weight calculations, or 3) data become sparse near the end of long-term follow-up in the control arm.

As demonstrated by Monte Carlo simulation studies in Section 3.3, when the switching rate in the control arm is extremely high (>85%), bias, type I error rate, and 95% coverage rate will significantly veer from nominal levels due to random non-positivity for both MSM and IPCW. The same is true if

the switching rate in the control arm is extremely low ($<15\%$). Therefore, for RCTs with a one-way switching design, we recommend avoiding using these two re-weighting-based approaches when the treatment switching rate is lower than 20% or higher than 80%.

To reduce the random non-positivity associated with including too many unnecessary covariates in the pooled logistic regression models for deriving inverse probability weights, it is essential to pre-specify in the SAP the criterion for selecting only covariates that are truly associated with treatment switching or censoring distributions. For example, excluding covariates with a $p\text{-value} = >0.2$. Priorities should be set according to the strength of (suspected) causal relationships between a covariate and the outcome, as small imbalances in a strong predictor of the outcome may cause more severe (residual) confounding bias than large differences in covariates that are only weakly related to the outcome.

In addition, it is also necessary to prevent individual patients from contributing too much information to the causal inference of randomized treatment effects. For this purpose, weight truncation criteria need to be pre-defined in the SAP and strictly followed. For example, reset all extremely large weights randomly to less than 2 times the 99th percentile of the weight distribution.

A necessary condition for well-behaved weight model specification is that the stabilized weights are close to a mean of 1 (Hernán et al. 2001). After considering all the factors discussed above, the final weight model chosen is a bias-variance trade-off between the inclusion of a sufficient number of confounders and the construction of a well-behaved model that has a small effect estimate variance. Refer to Hernán et al. (2001) for an insightful example on how to settle on parameterizing weight models.

4.3. Correct model specifications

As discussed in Section 2, for MSM analysis in RCTs with a one-way treatment switching setting, there are two types of known model misspecifications: 1) introducing an R by A(t) interaction term in Model (6) or (7); and 2) omitting baseline confounding factors in the final MSM model when they are not balanced between randomized arms and when stabilized weights are used. Therefore, in the SAP, measures to prevent these two types of misspecifications need to be pre-planned.

Estimated weights with very extreme values or mean weights far from 1 are indicative of misspecification in models for inverse probability weight calculations. If the necessary conditions for correct model specification are violated (Cole and Hernán 2008), it is important to check whether any key time-dependent confounders have been left out of the weight calculation models or whether the weight models are not correctly specified in other ways.

The general MSM specification in Section 2 for RCTs with a one-way treatment switching setting includes an assumption for treatment switching in the style of an ITT approach – i.e., A(t) is continuously used without change after it is initiated by a control arm patient. However, in RCTs, this assumption may often be violated. For some patients, after A(t) is initiated, it may be stopped early or a different type or multiple types of A(t) may be used, which may make the correct causal inference impossible for a particular alternative treatment effect. However, this will not affect the valid causal inference for the randomized treatment effect, which is the primary focus of MSM's application in RCTs.

5. A case study

For illustrative purposes, the proposed MSM approach was applied to case study data derived from a subset of patients from a Takeda phase 3 randomized trial on multiple myeloma. The original trial design permits patients to switch to multiple alternative treatments after relapse, including taking medications that belong to the same class as the active drug. Patients in the placebo arm switching to the same active medication class are treated as one-way treatment switching in this case study. All

treatment switching patients in the placebo arm and randomly selected patients without treatment switching (excluding those switching to other class(es) of alternative therapies) in both arms are used in this case study. The final case study dataset consists of 244 patients, with 121 patients in the active arm and 123 patients in the placebo arm with 89 experiencing treatment switching. Fifty patients died in the placebo arm, out of whom 30 were without treatment switching; 54 died in the active arm. Patients were followed for up to 70 months in this dataset.

In this case study, progression-free survival (PFS) is in favor of patients in the active arm versus the placebo ($HR_{PFS} = 0.553$, $p\text{-value} < 0.001$, and $95\% CI_{PFS} = [0.400, 0.763]$). Based on an ITT analysis, the PFS benefit did not fully translate into an OS benefit for the active intervention ($HR_{OS,ITT} = 0.948$, $p\text{-value} = 0.7856$, and $95\% CI_{OS,ITT} [0.649, 1.382]$). One-way treatment switching is one of the factors potentially impacting causal inference on the OS benefit. To adjust for possible time-dependent confounding associated with one-way treatment switching, we apply MSM to this dataset.

For the MSM analysis, Cox proportional hazards regression approximated via pooled logistic models (D'Agostino et al. 1990) was used to generate the probability of treatment switching for the placebo arm or censoring for both arms. Ten time-dependent covariates such as progression status, platelet count, and hemoglobin level and seven baseline covariates such as cytogenetics population group, baseline hemoglobin level, baseline LDH, and baseline platelets were kept in the final four models for deriving stabilized IPTWs and IPCWs from placebo arm data. For the active arm, IPTWs were set as 1 for all patients at all observed time points and IPCWs were derived by calculating the probability of censoring through pooled logistic models where the same time-dependent and baseline covariates were included in the models as for the control arm.

Figure 1 summarizes the $SW(t)$ distributions by study quarter. The weight distributions skew towards less than one over time, but both means and medians (as denoted by green diamonds and thick horizontal bars in the figure, respectively) are around 1 at each quarter. The overall mean of $SW(t)$ at all study visits is 0.99, with a standard deviation of 0.24. Within the control arm, the weight range post-trimming is 0.01 to 5.1; these weights were derived from treatment switching and informative censoring distributions and expected to make a large adjustment. Within the active

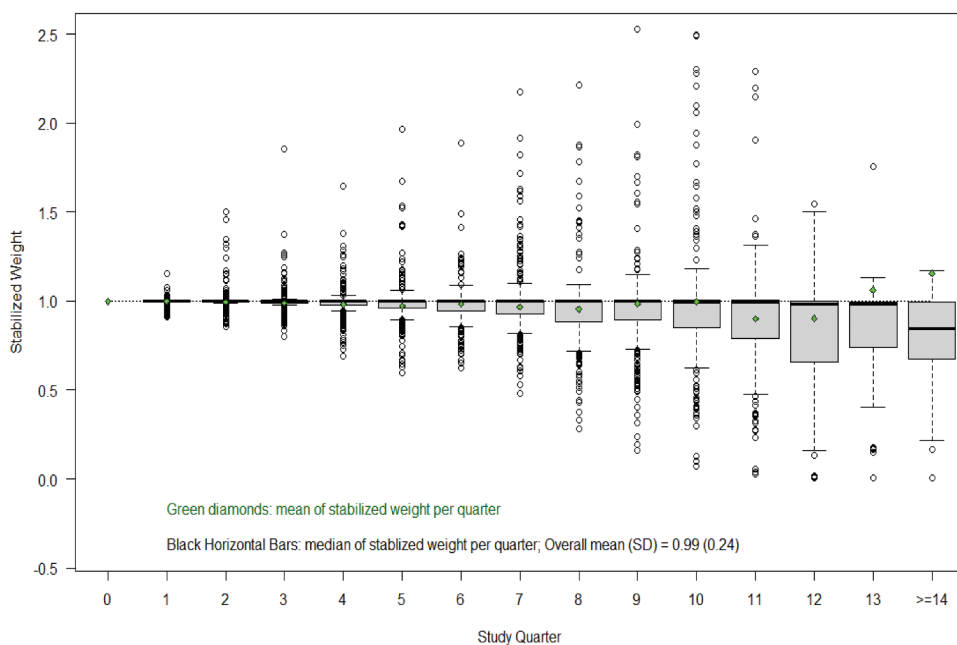


Figure 1. Case study data: stabilized weight distribution by study quarter.

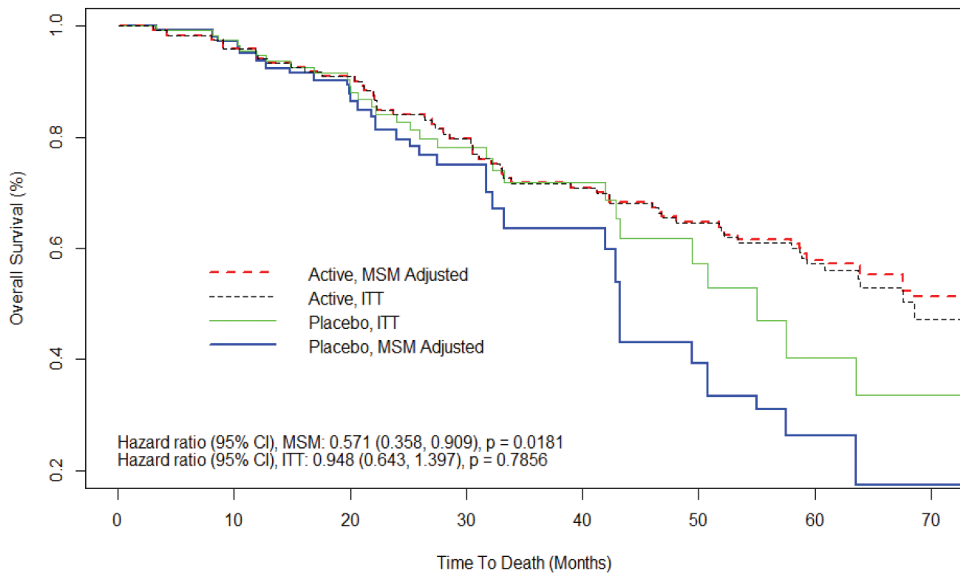


Figure 2. Case study data: overall survival outcome: ITT versus MSM.

arm, the weight range post-trimming is 0.24 to 2.2; these weights were derived from the informative censoring distribution and expected to make a small adjustment.

Figure 2 compares the OS curves derived from the ITT analysis and the MSM adjusted analysis via the Adjusted Kaplan – Meier Estimator (AKME) (Xie and Liu 2005; Xu et al. 2022). Looking at the MSM adjusted curves, if there had been no treatment switching the survival probability for the placebo patients would have been lower, and the survival probability for patients in the active arm would have been slightly higher in the late stage of the trial if no informative censoring had occurred. The median difference between control and active survival curves increased post MSM adjustment. The estimated active treatment adjusted $HR_{(OS, MSM, R)} = 0.571$, p-value $_{(OS, MSM, R)} = 0.0181$, and $95\% CI_{(OS, MSM, R)} = [0.358, 0.909]$. From sensitivity analysis, where no trimming was exercised, the MSM curves remain similar.

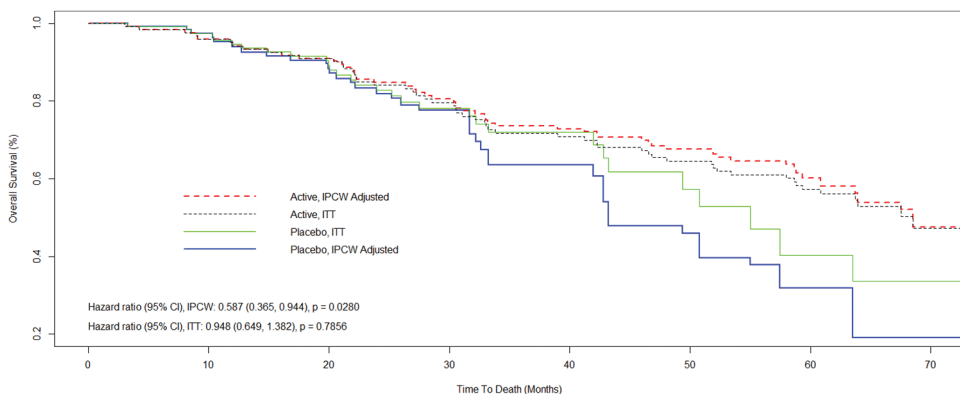


Figure 3. Case study data: overall survival outcome: ITT versus IPCW.

As a by-product, from MSM, the estimated switched-treatment effect $HR_{(OS, MSM, A)} = 0.236$, p -value $_{(OS, MSM, A)} < 0.0001$, 95% $CI_{(OS, MSM, A)} = [0.123, 0.453]$, indicating that if the active agent had been used as the next line treatment on the relapsed patients, it could have been effective. The limitation of $HR_{(OS, MSM, A)}$ outcome: the result is based on a small sample size and a small number of cases from this case study dataset (25 OS events among 89 treatment-switched patients).

To validate the MSM approach outcome, the case study data were analyzed via IPCW as well. The adjusted analysis result from IPCW is summarized in Figure 3, where less downward adjustment occurred for the placebo arm survival curve as compared to the ITT approach, but more upward adjustment occurred for the active treatment arm than with MSM. This is due to different model specifications between MSM and IPCW. Nevertheless, the IPCW outcome is very similar to MSM, with an estimated adjusted $HR_{(OS, IPCW, R)} = 0.587$, p -value $_{(OS, IPCW, R)} = 0.0280$, and 95% $CI_{(OS, IPCW, R)} = [0.365, 0.944]$.

6. Conclusion and discussion

In this article, we demonstrated that MSM can be used for causal inference when analyzing RCT data having a one-way treatment switching design, but with different model specifications than used in observational studies and in RCTs where both arms allow treatment switching. Special attention needs to be paid to avoid structural non-positivity. Monte Carlo Simulation results demonstrated that if MSM is used appropriately in analyzing RCT data with one-way treatment switching, the bias and type I error will be well controlled when the treatment switching rate is not too low or too high in the control arm. The simulation outcomes indicate that when the one-way treatment switching rate is less than 15% or more than 85%, bias may increase substantially when using the MSM approach, and the type I error may exceed the pre-specified nominal level. This is due to the elevated random non-positivity rate when the one-way treatment switching rate is too far away from 50%. MSM is appealing because it is intuitive and is a natural extension of the time-dependent Cox model. By using AKME (Xie and Liu 2005) and the Cox partial likelihood estimator jointly, outputs from MSM resemble those from the Cox proportional-hazards regression model and Kaplan–Meier analysis, respectively. Thus, it is easy to communicate MSM results to clinicians and non-statisticians.

Application of MSM in RCTs with a one-way treatment switching setting needs to meet the unmeasured confounding assumption, which is not testable. To accommodate this assumption adequately, it is important to make sure that a rich collection of prognostic factors be specified for collection during the clinical trial planning stage – i.e., all major potential confounding variables, both baseline and time dependent, that may affect disease progression, treatment switching, and/or death should be identified in the trial design stage through feedback from clinicians and key opinion leaders (KOL's) and based on intensive literature review. Additionally, these potential confounders need to be prespecified in the SAP as covariate candidates for the MSM analysis.

In some RCTs allowing for treatment switching, in addition to one-way treatment switching from control to active arm, some patients in both arms may change to alternative interventions outside of the study's investigational products after disease progression. One limitation of the method proposed in this manuscript is that it cannot make causal inference on randomized active intervention treatment effects under this more complicated situation – this is a topic we will pursue in future research.

MSM and IPCW analyses may reuse the same counting process dataset. However, IPCW makes an additional assumption not present in MSM: uncensored patients must represent censored patients well after weighting. If the assumption does not hold well, under- or over-adjustment may occur via IPCW. Adjusted OS results may be more reliable if multiple causal inference approaches yield similar results. Therefore, in the SAP, it is better to pre-specify at least two causal inference approaches for OS adjusted analyses if the trial has a one-way treatment switching design.

MSM, TSE, RPSFTM, and IPCW are powerful tools that may uncover a causal effect for active treatment on OS that is otherwise obscured by one-way treatment switching in the control arm. In practice, the performance of each method will be data driven and no single method may consistently out-perform others under different situations; powerful tools can be dangerous if not handled with care (Cole and Hernán 2008). As shown in Section 4 and in our previous manuscript (Xu et al. 2022), bias may be introduced and the type I error may be inflated if consistency, positivity, and correct model specification requirements are not followed carefully.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- Addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials E9(R1). 2019. International Council for Harmonisation (ICH), Geneva, Switzerland.
- Branson, M., and J. Whitehead. 2002. Estimating a treatment effect in survival studies in which patients switch treatment. *Statistics in Medicine* 21 (17):2449–2463. doi: [10.1002/sim.1219](https://doi.org/10.1002/sim.1219).
- Cole, S. R., and M. A. Hernán. 2008. Constructing inverse probability weights for marginal structural models. *American Journal of Epidemiology* 168 (6):656–664. doi: [10.1093/aje/kwn164](https://doi.org/10.1093/aje/kwn164).
- D’Agostino, R. B., M. L. Lee, A. J. Belanger, L. A. Cupples, K. Anderson, and W. B. Kannel. 1990. Relation of pooled logistic regression to time dependent Cox regression analysis: The Framingham Heart Study. *Statistics in Medicine* 9 (12):1501–1515. doi: [10.1002/sim.4780091214](https://doi.org/10.1002/sim.4780091214).
- Farmer, R. E., D. Kounali, A. S. Walker, J. Savović, A. Richards, M. T. May, and D. Ford. 2018. Application of causal inference methods in the analyses of randomised controlled trials: A systematic review. *Miomed Central* 19 (23):1–14. doi: [10.1186/s13063-017-2381-x](https://doi.org/10.1186/s13063-017-2381-x).
- Hernán, M. A., B. Brumback, and J. M. Robins. 2001. Marginal structural models to estimate the joint causal effect of nonrandomized treatments. *Journal of the American Statistical Association* 96 (454):440–448. doi: [10.1198/016214501753168154](https://doi.org/10.1198/016214501753168154).
- Hernán, M. Á., B. Brumback, and J. M. Robins. 2000. Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men. *Epidemiology* 11 (5):561–570. doi: [10.1097/00001648-200009000-00012](https://doi.org/10.1097/00001648-200009000-00012).
- Latimer, N. R., K. R. Abrams, P. C. Lambert, M. J. Crowther, A. J. Wailoo, J. P. Morden, R. L. Akehurst, and M. J. Campbell. 2017. Adjusting for treatment switching in randomised controlled trials – a simulation study and a simplified two-stage method. *Statistical Methods in Medical Research* 26 (2):724–751. doi: [10.1177/0962280214557578](https://doi.org/10.1177/0962280214557578).

- Latimer, N. R., K. R. Abrams, and U. Siebert. 2019. Two-stage estimation to adjust for treatment switching in randomised trials: A simulation study investigating the use of inverse probability weighting instead of re-censoring. *BMC Medical Research Methodology* 19 (1):69. [10.1186/s12874-019-0709-9](https://doi.org/10.1186/s12874-019-0709-9).
- Rimawi, M., and S. G. Hilsenbeck. 2012. Making sense of clinical trial data: Is inverse probability of censoring weighted analysis the answer to crossover bias. *Clinical Oncology* 30 (4):453–458. doi: [10.1200/JCO.2010.34.2808](https://doi.org/10.1200/JCO.2010.34.2808).
- Robins, J. M. 2000. *Marginal structural models versus structural nested models as tools for causal inference*. New York: Springer.
- Robins, J. M., and D. M. Finkelstein. 2000. Correcting for noncompliance and dependent censoring in an AIDS clinical trial with inverse probability of censoring weighted (IPCW) log-rank tests. *Biometrics* 56 (3):779–788. doi: [10.1111/j.0006-341X.2000.00779.x](https://doi.org/10.1111/j.0006-341X.2000.00779.x).
- Robins, J. M., M. A. Hernán, and B. Brumback. 2000. Marginal structural models and causal inference in epidemiology. *Epidemiology* 11 (5):550–560. doi: [10.1097/00001648-200009000-00011](https://doi.org/10.1097/00001648-200009000-00011).
- Robins, J. M., and A. A. Tsiatis. 1991. Correcting for non-compliance in randomized trials using rank preserving structural failure time models. *Communications in Statistics-Theory and Methods* 20 (8):2609–2631. doi: [10.1080/03610929108830654](https://doi.org/10.1080/03610929108830654).
- Rotnitzky, A., and J. Robins. 2005. Inverse probability weighted estimation in survival analysis. *Encyclopedia of Biostatistics* 4:2619–2625.
- SAS Institute Inc. 2011. *SAS/ETS 9.1 user's guide*. Cary, NC: SAS Institute Inc.
- Xie, J., and C. Liu. 2005. Adjusted Kaplan–Meier estimator and log-rank test with inverse probability of treatment weighting for survival data. *Statistics in Medicine* 24 (20):3089–3110. <https://onlinelibrary.wiley.com/doi/abs/10.1002/sim.2174>.
- Xu, J., G. Liu, and B. Wang. 2022. Bias and type I error control in correcting treatment effect for treatment switching using marginal structural models in phase III oncology trials. *Journal of Biopharmaceutical Statistics* 32 (6):1–18. doi: [10.1080/10543406.2022.2058524](https://doi.org/10.1080/10543406.2022.2058524).
- Yamaguchi, T., and Y. Ohashi. 2004. Adjusting for differential proportions of second-line treatment in cancer clinical trials. Part I: Structural nested models and marginal structural models to test and estimate treatment arm effects. *Statistics in Medicine* 23 (13):1991–2003. doi: [10.1002/sim.1816](https://doi.org/10.1002/sim.1816).
- Young, J. G., and E. J. Tchetgen Tchetgen. 2014. Simulation from a known Cox MSM using standard parametric models for the g-formula. *Statistics in Medicine* 33 (6):1001–1014. doi: [10.1002/sim.5994](https://doi.org/10.1002/sim.5994).